

CUSTOMER PERSONALITY ANALYSIS

Exploratory Data Analysis Report

A Comprehensive Study of Customer Demographics, Spending Behavior,
Purchasing Patterns, and Marketing Campaign Performance

Dataset: customer_personality_cleaned.csv | 2,240 Customer Records | 33 Variables

Prepared: July 2026

Executive Summary

This report presents a complete exploratory data analysis (EDA) of a customer personality and marketing dataset covering 2,240 customers across 33 variables. The analysis examines customer demographics, spending behavior, purchasing channels, website engagement, recency, marketing campaign performance, complaints, correlations between key metrics, and customer segmentation, concluding with actionable business insights and strategic recommendations.

Key Insights

- The dataset is clean: zero duplicate records and zero missing values across all 33 columns.
- Average customer spending is \$568, but spending is highly skewed — the top customer spent \$2,099.50 versus a minimum of \$5.00.
- Wine and Meat products dominate revenue, together accounting for roughly 80% of total spending.
- Income shows a strong positive correlation with spending ($r = 0.80$) and purchase frequency ($r = 0.75$).
- Store purchases remain the leading sales channel, ahead of web and catalog purchases.
- 38.5% of customers (862) have been inactive for over 60 days and are prime candidates for re-engagement campaigns.
- High-value customers make up 25% of the customer base and should be prioritized for loyalty and retention programs.
- The AcceptedCmp1–5, Response, and Complain fields contain only zero values in this dataset extract — a data quality issue flagged in Tasks 7, 8, and 11 that should be resolved before drawing campaign- or complaint-specific conclusions.

Task 1 — Dataset Overview

The dataset was loaded from `customer_personality_cleaned.csv` and verified for structural integrity before proceeding with analysis.

1.1 Dataset Dimensions

Metric	Value
Total Records (Rows)	2,240
Total Attributes (Columns)	33

1.2 Column Names

The dataset contains the following 33 columns:

ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Dt_Customer, Recency, MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds, NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5, AcceptedCmp1, AcceptedCmp2, Complain, Z_CostContact, Z_Revenue, Response, Age, Year, Month, Day.

1.3 Data Types

Two categorical (text) fields — `Education` and `Marital_Status` — plus a date field (`Dt_Customer`). All remaining core numeric fields (`Income`, `spending`, `purchase`, and `campaign` columns) are stored as floating-point numbers; `Year`, `Month`, and `Day` (derived from `Dt_Customer`) are integers.

1.4 Descriptive Statistics (Selected Fields)

Field	Mean	Std Dev	Min	25%	Median	75%	Max
Income	51,865.87	20,936.10	1,730	35,538.75	51,381.50	68,289.75	117,416.25
Age	45.15	11.77	18	37	44	55	82
Recency (days)	49.11	28.96	0	24	49	74	99
MntWines	302.11	331.06	0	23.75	173.50	504.25	1,225
MntMeatProducts	150.87	178.82	0	16	67	232	556

1.5 Duplicate Records

A full-row duplicate check was performed using `df.duplicated()`. Result: no duplicate records were found in the dataset.

1.6 Missing Values

A null-value audit was run across all 33 columns using `df.isnull().sum()`. Result: every column returned zero missing values — the dataset is fully complete and required no imputation.

Key Insights

- Dataset is production-ready: 2,240 rows × 33 columns, no duplicates, no missing values.
- Income and spending fields are right-skewed, with high-value outliers pulling the mean above the median.
- Age ranges from 18 to 82, with a median of 44 — a broad but middle-aged-centered customer base.

Task 2 — Customer Demographic Analysis

This section profiles customers by age, income, education, marital status, and household composition.

2.1 Age & Income Distribution

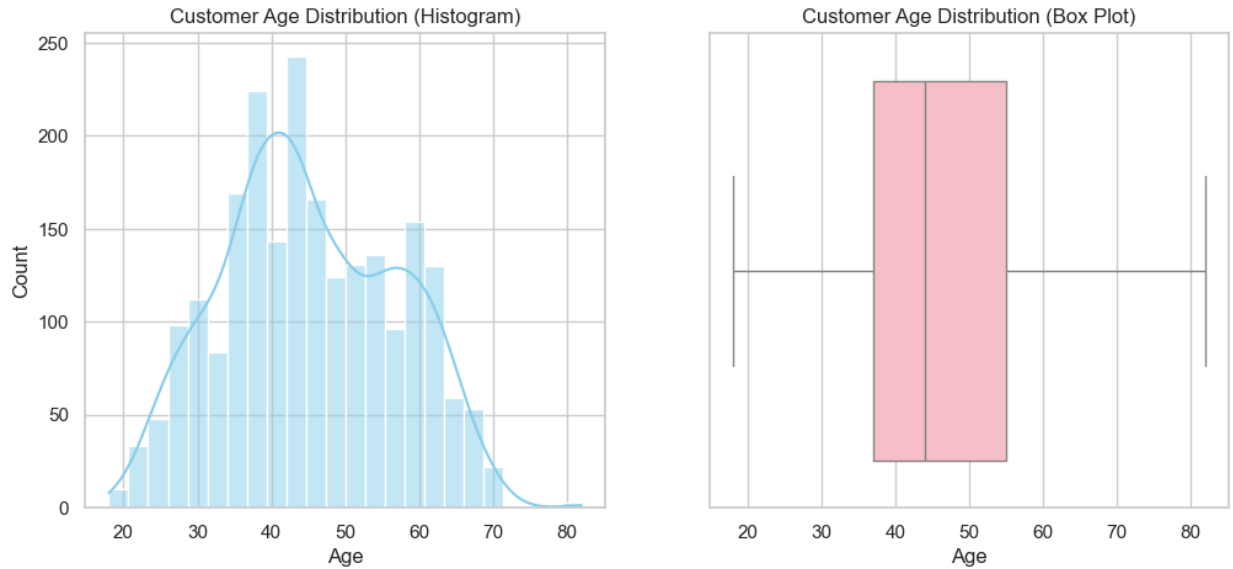


Figure 1. Customer age distribution — histogram (left) and box plot (right).

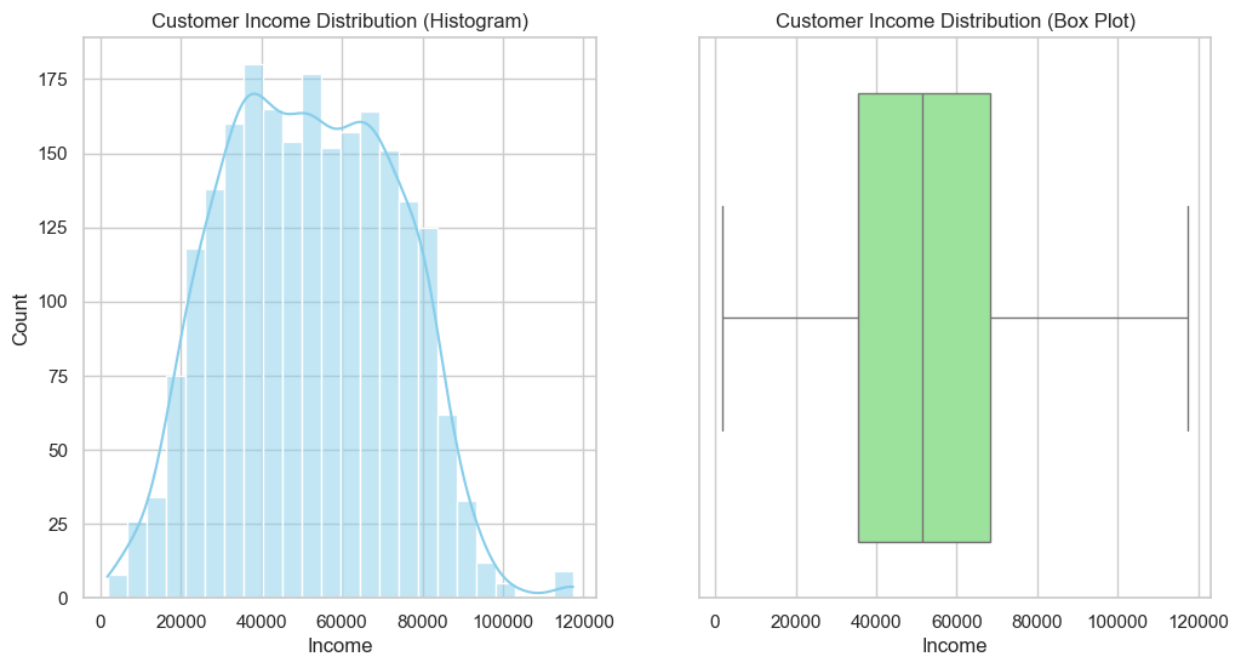


Figure 2. Customer income distribution — histogram (left) and box plot (right).

Customer age ranges from 18 to 82 years, with a median age of 44. Median household income is \$51,381.50, with a right-skewed distribution and a small number of high-income outliers.

2.2 Education & Marital Status

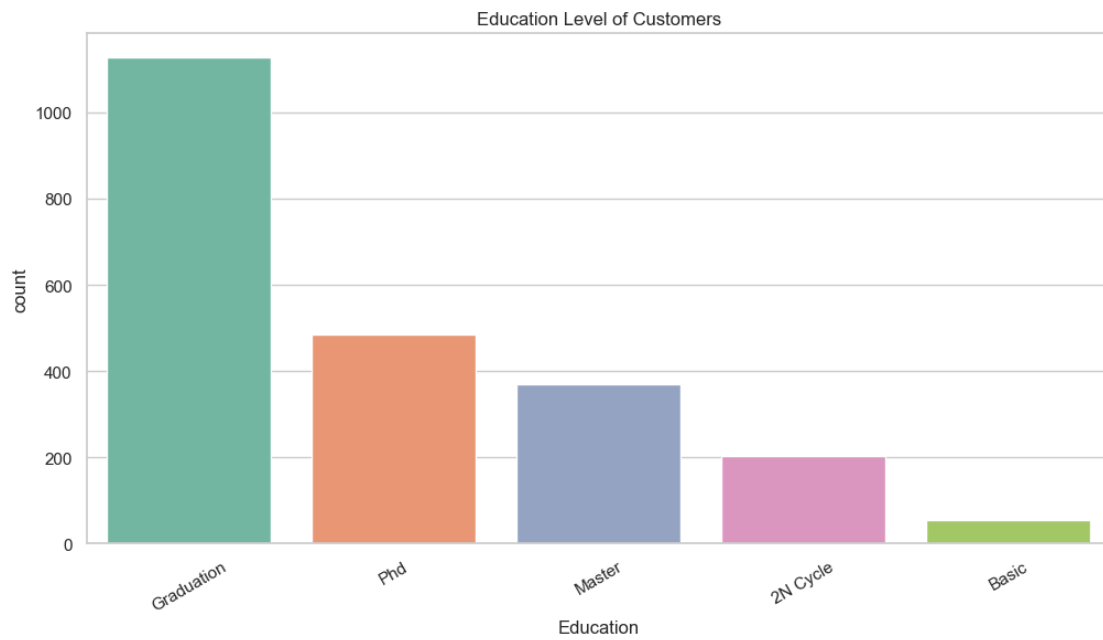


Figure 3. Distribution of customers by education level.

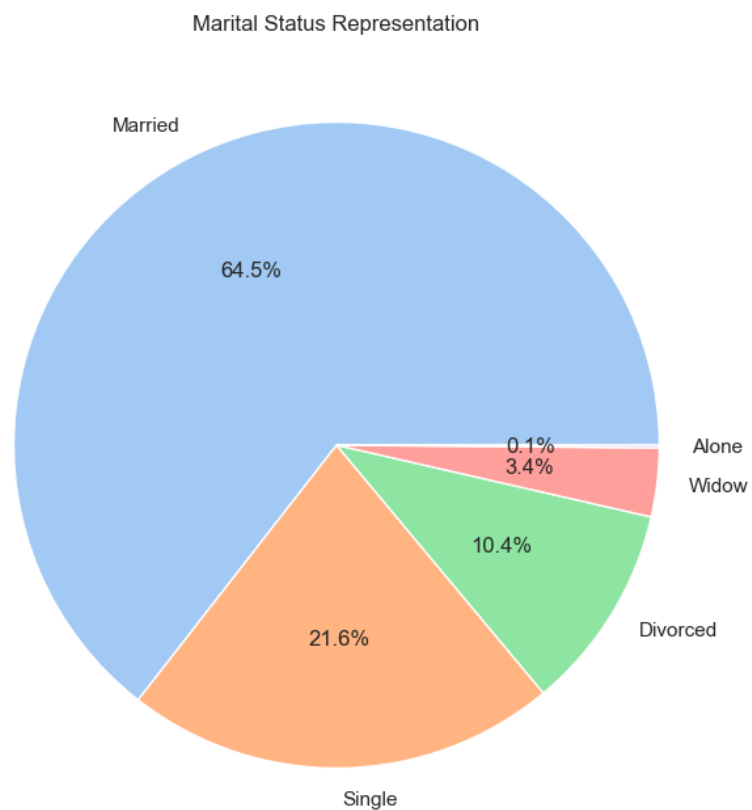


Figure 4. Marital status representation (pie chart).

"Graduation" is the most common education level, and "Married" is the most common marital status among customers.

2.3 Household Composition

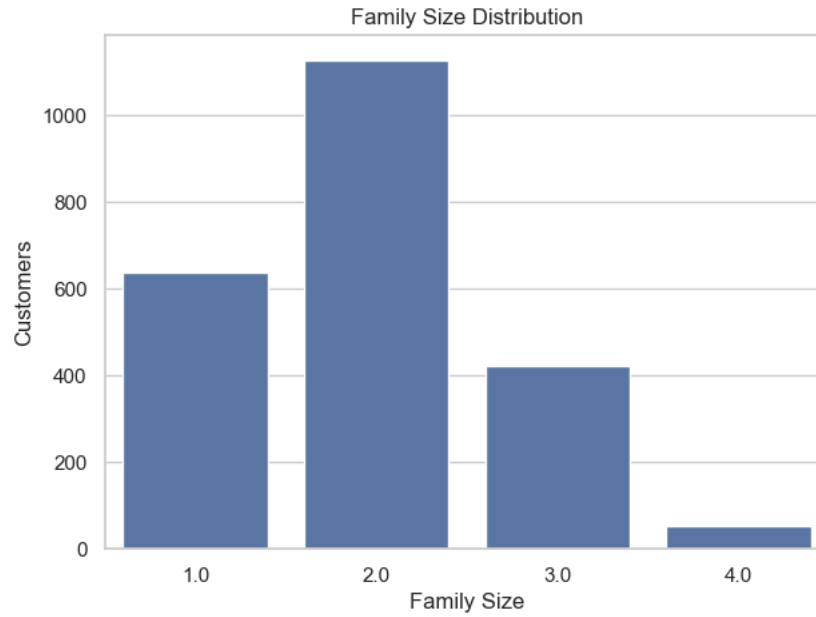


Figure 5. Distribution of family size (1 = single, no dependents; up to 4).

Family Size	Number of Customers
2 (couple, no dependents)	1,128
1 (single, no dependents)	638
3	421
4	53

2.4 Number of Children

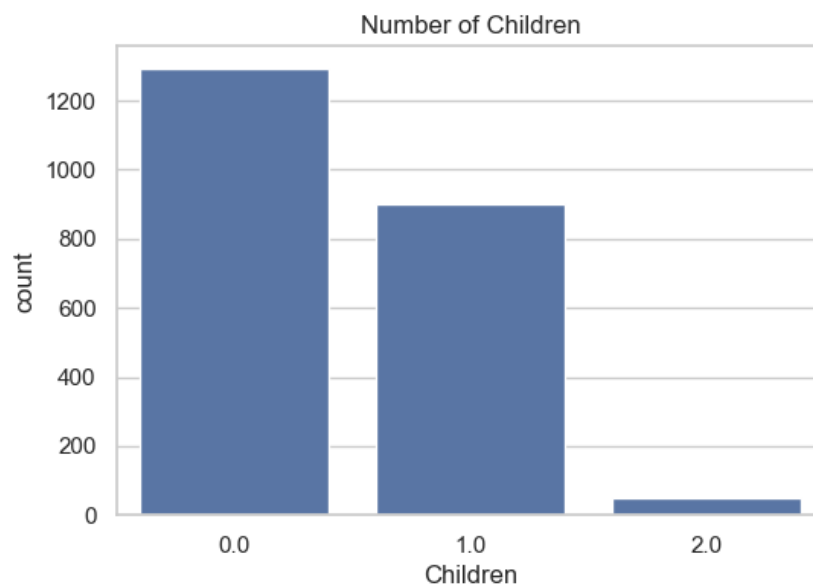


Figure 6. Number of children (Kidhome) per household.

Children at Home	Customers
0	1,293
1	899
2	48

2.5 Number of Teenagers

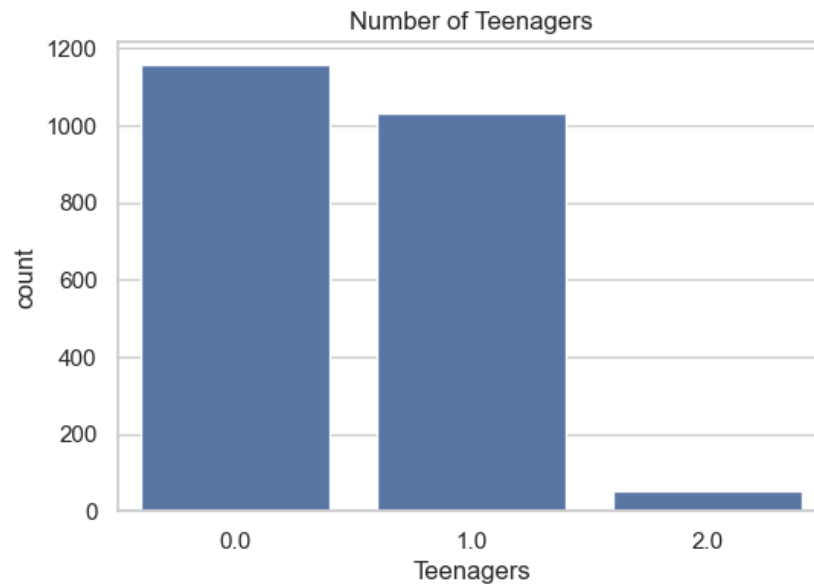


Figure 7. Number of teenagers (Teenhome) per household.

Teenagers at Home	Customers
0	1,158
1	1,030
2	52
Key Insights Average customer age is 45.1 years; average household income is \$51,865.87.	

Teenagers at Home	Customers
• Most customers hold a Graduation-level education and are Married. • Average family size is 1.95 — most households are couples or singles with limited dependents. • On average, customers have 0.44 children and 0.51 teenagers at home, so most households (57.7%) have zero children.	

Task 3 — Customer Spending Analysis

A Total_Spending field was engineered by summing all six product-category spending fields (Wine, Fruits, Meat, Fish, Sweets, Gold) for each customer.

3.1 Overall Spending Summary

Metric	Value (\$)
Total Spending (all customers)	1,272,783.00
Average Spending per Customer	568.21
Maximum Spending (single customer)	2,099.50
Minimum Spending (single customer)	5.00

3.2 Highest & Lowest Spending Customers

Attribute	Highest Spender (ID 4580)	Lowest Spender (ID 11110)
Total Spending	\$2,099.50	\$5.00
Income	\$75,759	\$3,502
Age	45	41
Education / Marital Status	Graduation / Married	Graduation / Single
Web / Catalog / Store Purchases	9 / 7 / 9	0 / 0 / 0

3.3 Spending by Product Category

Category	Total Spending (\$)	Average Spending (\$)	Share of Total
Wine	676,733.00	302.11	53.2%
Meat	337,956.00	150.87	26.6%
Gold Products	88,452.50	39.49	6.9%
Fish	71,939.50	32.12	5.7%
Sweets	49,141.00	21.94	3.9%
Fruits	48,561.00	21.68	3.8%

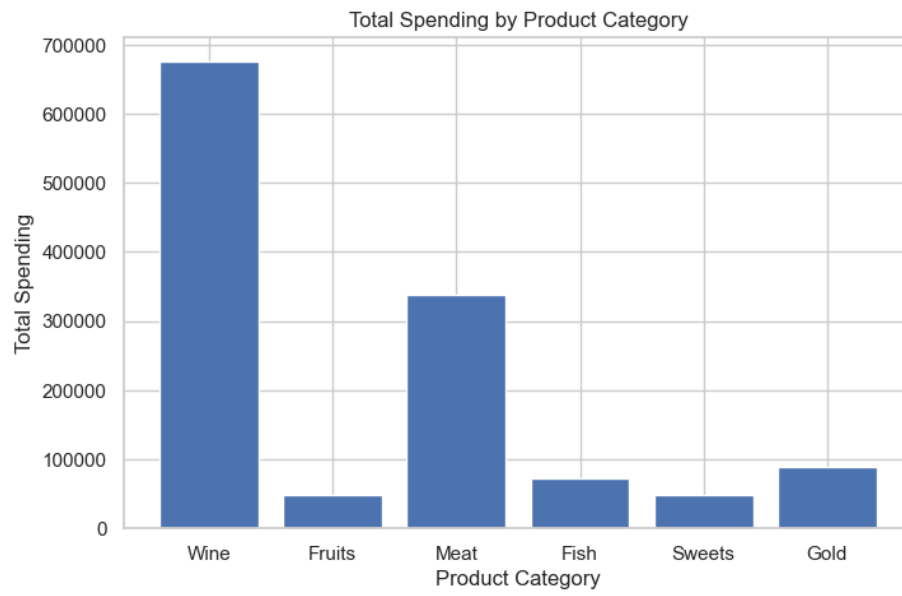


Figure 8. Total spending by product category (bar chart).

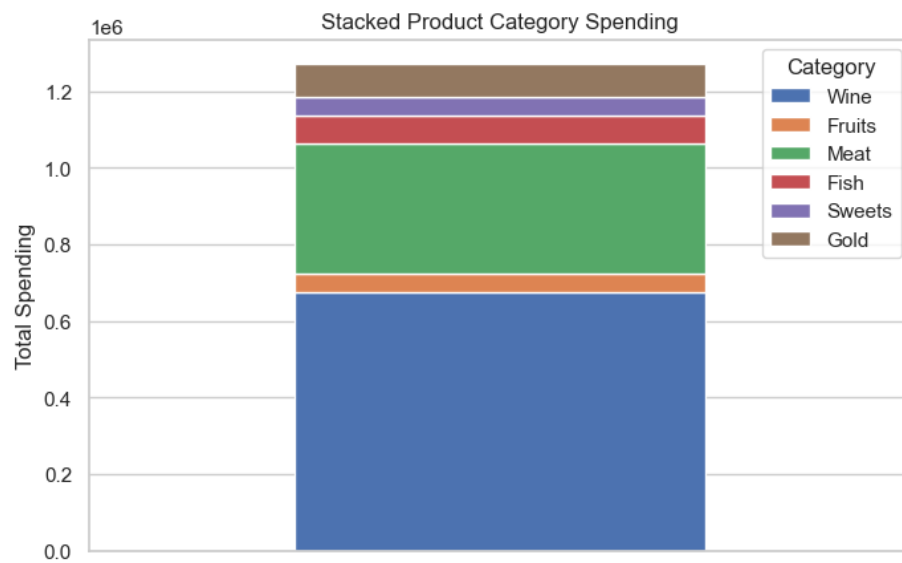


Figure 9. Stacked bar chart of category contribution to total spending.

3.4 Distribution of Total Customer Spending

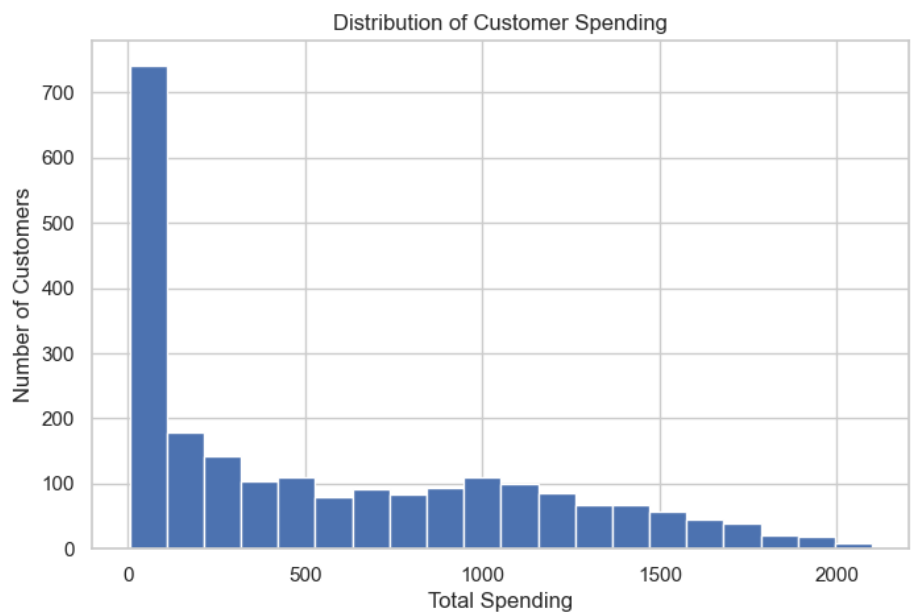


Figure 10. Histogram of total spending per customer.

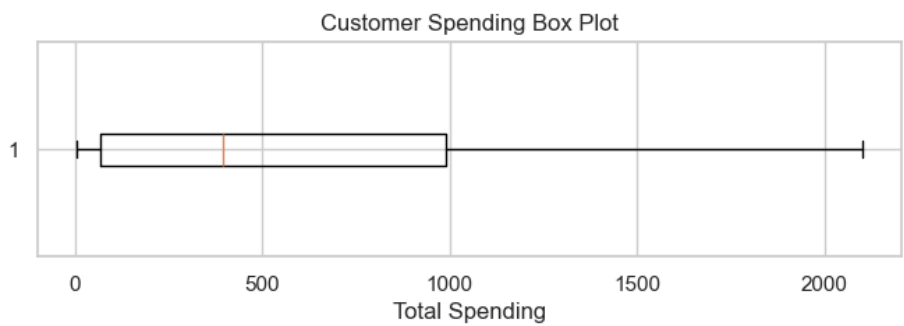


Figure 11. Box plot of total spending per customer.

3.5 Top 10 Highest-Spending Customers

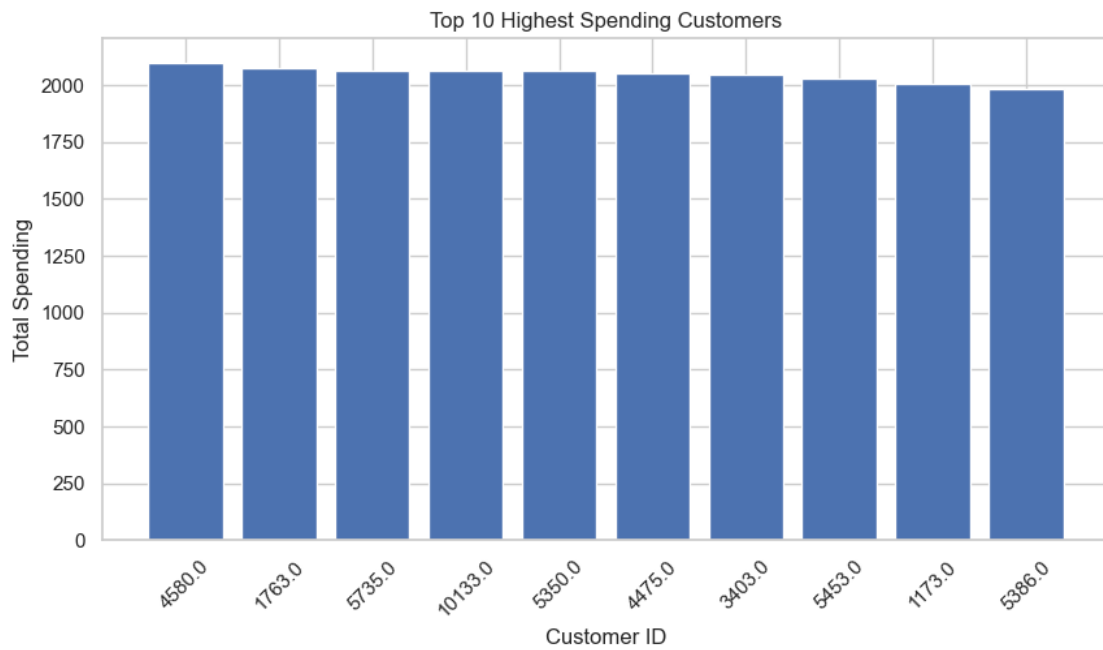


Figure 12. Top 10 customers ranked by total spending.

Key Insights

- Wine and Meat products together generate roughly 80% of all product revenue — the two priority categories for inventory and promotions.
- Spending is heavily right-skewed: a small group of high-value customers accounts for a disproportionate share of revenue.
- The gap between the highest (\$2,099.50) and lowest (\$5.00) spender is over 400x, reflecting very distinct customer value tiers.

Task 4 — Purchasing Behavior Analysis

Purchasing channels analyzed: in-store, web, and catalog, along with overall purchase frequency per customer.

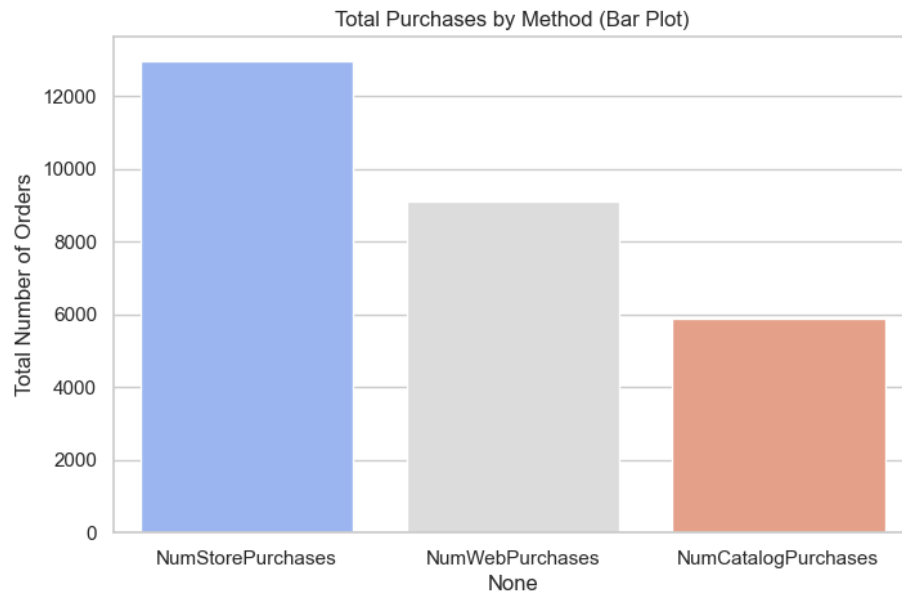


Figure 13. Total purchases by method (store, web, catalog).

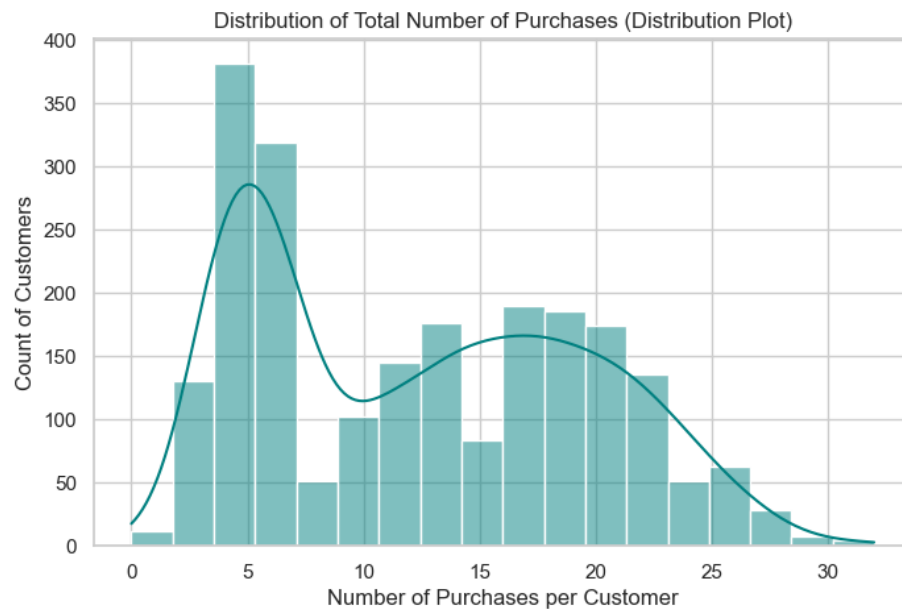


Figure 14. Distribution of total number of purchases per customer.

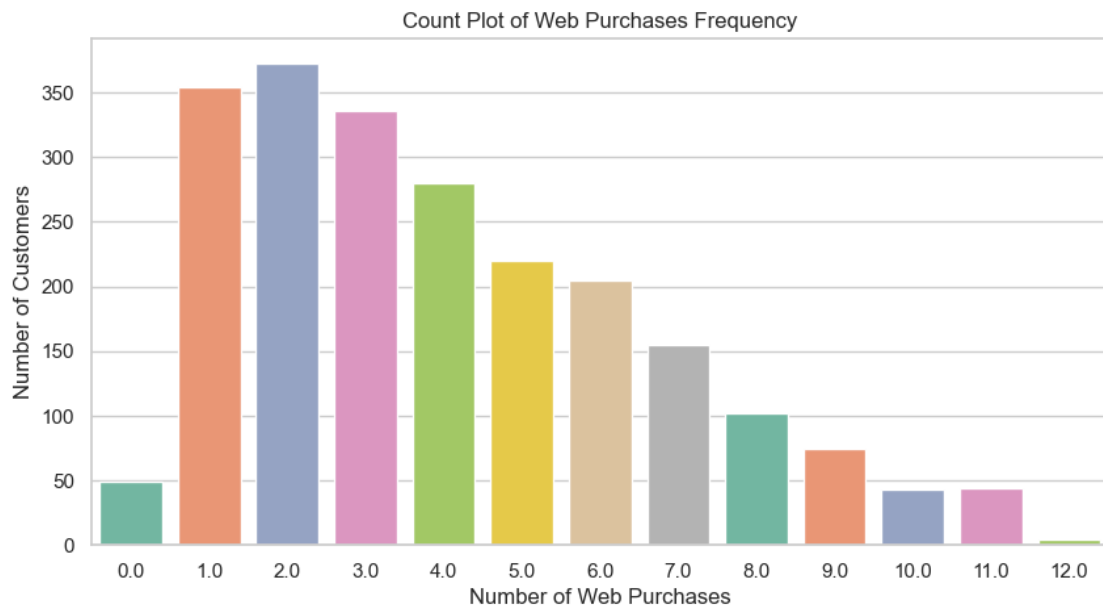


Figure 15. Count plot of web purchase frequency.

Key Insights

- In-store purchases (NumStorePurchases) are the leading purchase channel, ahead of web and catalog purchases.
- Purchase counts per customer are broadly distributed, indicating a mix of occasional and frequent buyers.
- Web purchase frequency is concentrated in the low-to-mid range (2–6 purchases), suggesting room to grow the online channel.

Task 5 — Website Engagement Analysis

Website engagement is measured via NumWebVisitsMonth, and cross-referenced against actual web purchases to gauge conversion.

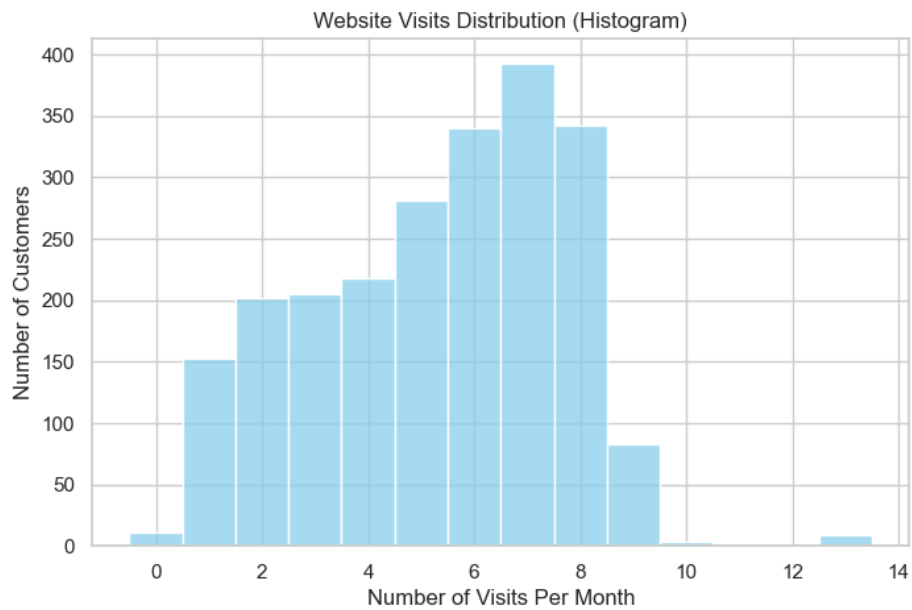


Figure 16. Distribution of website visits per month.



Figure 17. Website visits vs. web purchases (scatter plot).

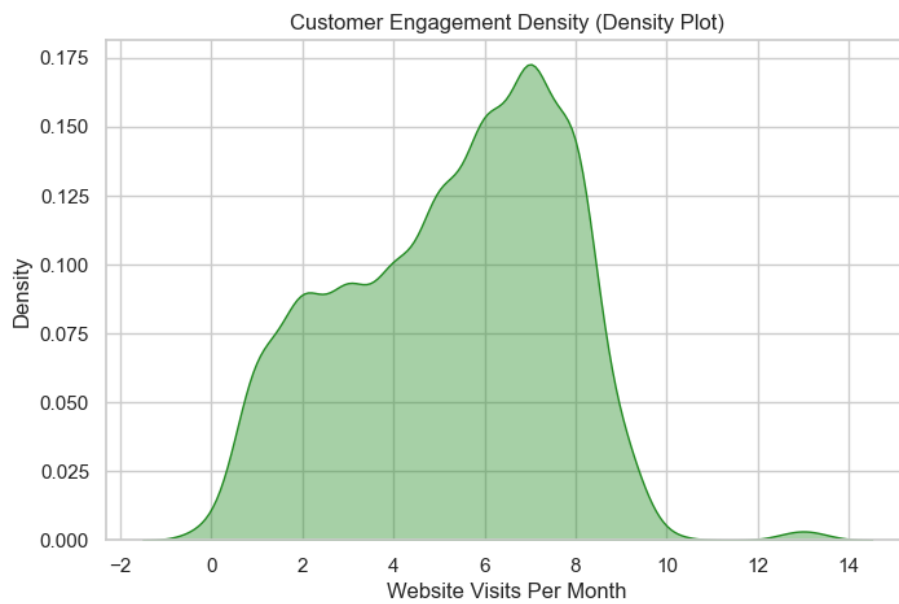


Figure 18. Density plot of customer engagement (website visits).

Key Insights

- The median customer visits the website 6 times per month.
- 437 customers are highly engaged, visiting the site more than 7 times per month.
- 366 customers show low engagement, visiting the site 2 times or fewer per month — a potential audience for re-engagement content or UX improvements.
- The scatter plot suggests visits alone do not guarantee purchases — browsing frequency and conversion are not perfectly aligned, meaning some frequent visitors are window-shoppers rather than buyers.

Task 6 — Customer Recency Analysis

Recency (days since last purchase) was used to classify customers into Active, Moderate, and Inactive segments using standard thresholds (≤ 30 days = Active, 31–60 = Moderate, > 60 = Inactive).

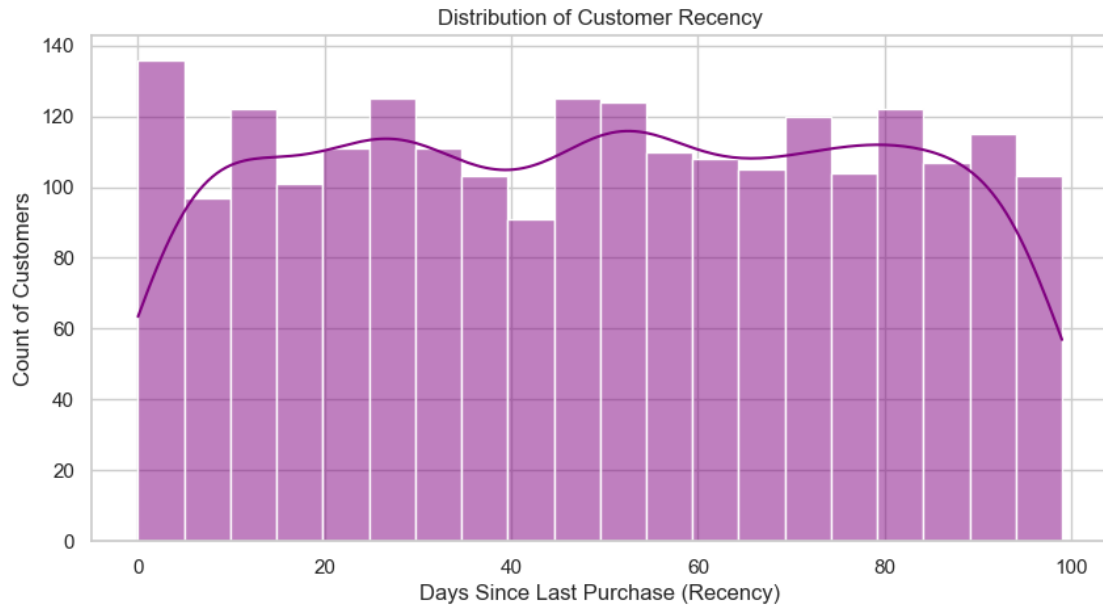


Figure 19. Distribution of customer recency (days since last purchase).

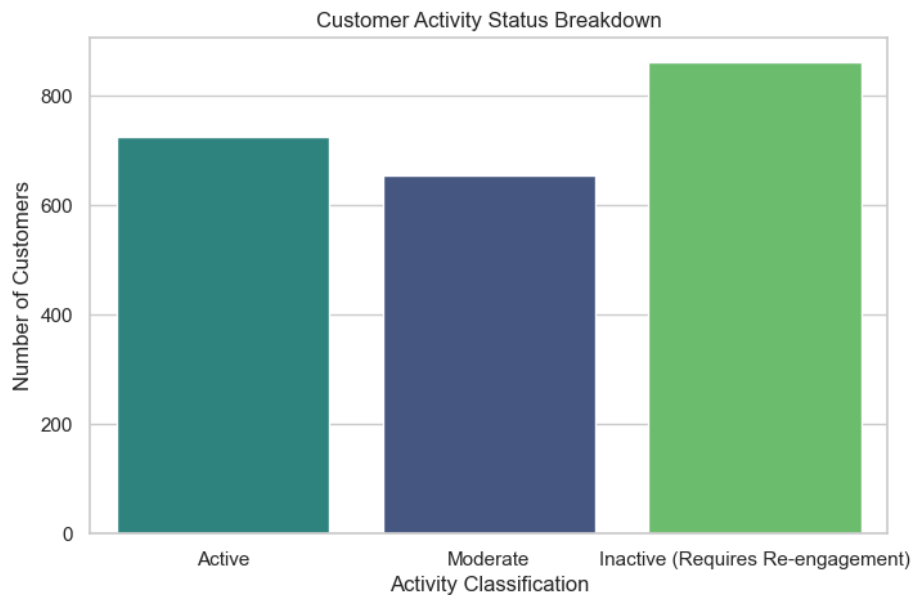


Figure 20. Customer activity status breakdown (Active / Moderate / Inactive).

Segment	Customers	Share
Active (≤ 30 days)	724	32.3%
Moderate (31–60 days)	654	29.2%

Segment	Customers	Share
Inactive (> 60 days)	862	38.5%
Key Insights - Recency is evenly spread from 0 to 99 days, indicating no strong recent purchasing spike or drop-off pattern. 862 customers (38.5%) have not purchased in over 60 days and represent the primary target for win-back / re-engagement campaigns. - Only 32.3% of customers are currently "Active," suggesting overall purchase cadence could be improved through more frequent		

Segment	Customers	Share
touchpoints or promotions.		

Business Insight: Re-engagement campaigns (targeted email offers, personalized discounts, or reminder communications) should be prioritized for the 862 inactive customers to recover lapsed revenue.

Task 7 — Marketing Campaign Analysis

Campaign acceptance was analyzed across five historical campaigns (Campaign 1–5) and the latest campaign (Response).

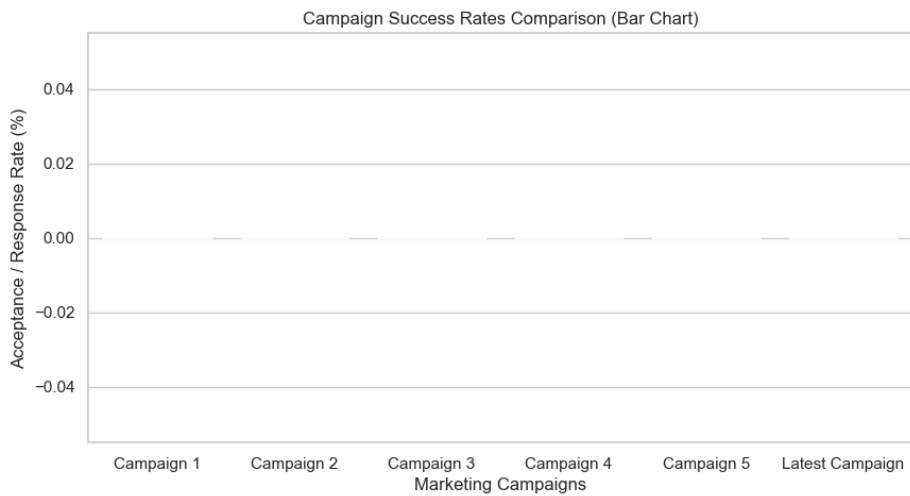


Figure 21. Campaign acceptance / response rate comparison.

Contribution Share of Each Campaign (No Data Available)

**No Campaigns Accepted (All Sums are 0.0)
Pie Chart Cannot Be Rendered**

Figure 22. Contribution share of each campaign.

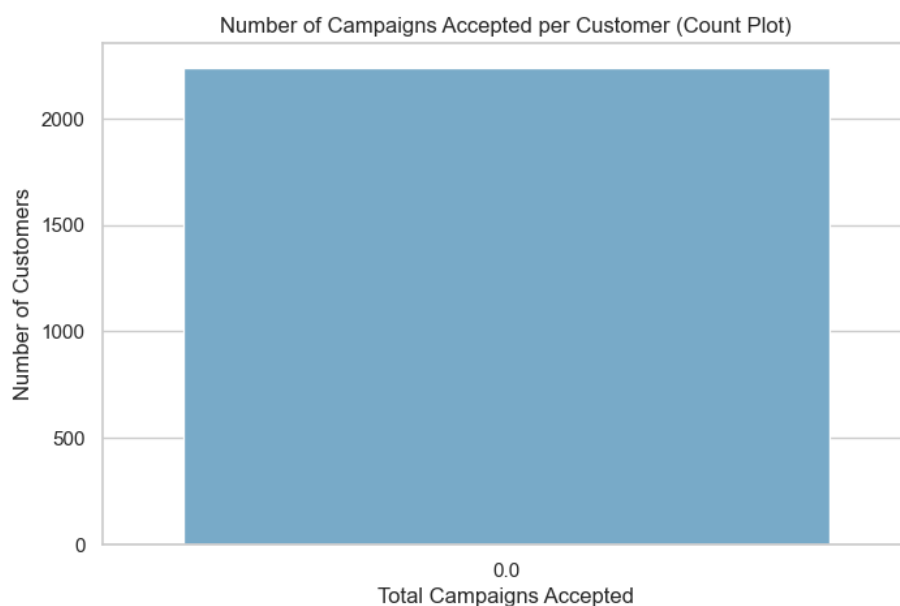


Figure 23. Number of campaigns accepted per customer.

Key Insights

- Data quality flag: in this dataset extract, the AcceptedCmp1, AcceptedCmp2, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5, and Response columns contain only zero values for all 2,240 customers — meaning no campaign acceptances or responses are recorded.
- As a result, acceptance rate, response rate, and "top performing campaign" cannot be meaningfully determined from this extract; all measured rates return 0.0%.
- Recommendation: verify the source extraction/ETL step that produced these columns, since a real-world marketing dataset would be expected to show non-zero acceptance for at least a subset of customers.

Task 8 — Customer Complaint Analysis

Complaint activity was analyzed using the Complain field, alongside its relationship to income and spending.

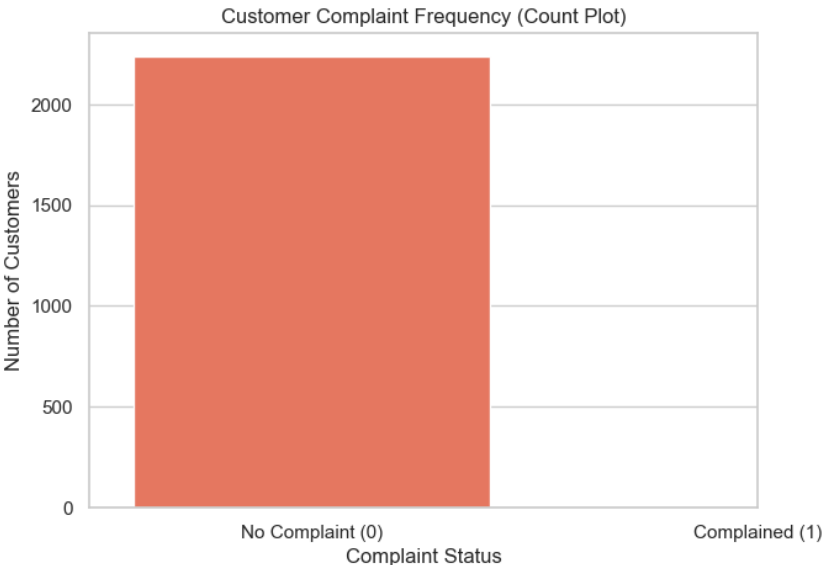


Figure 24. Customer complaint frequency (count plot).

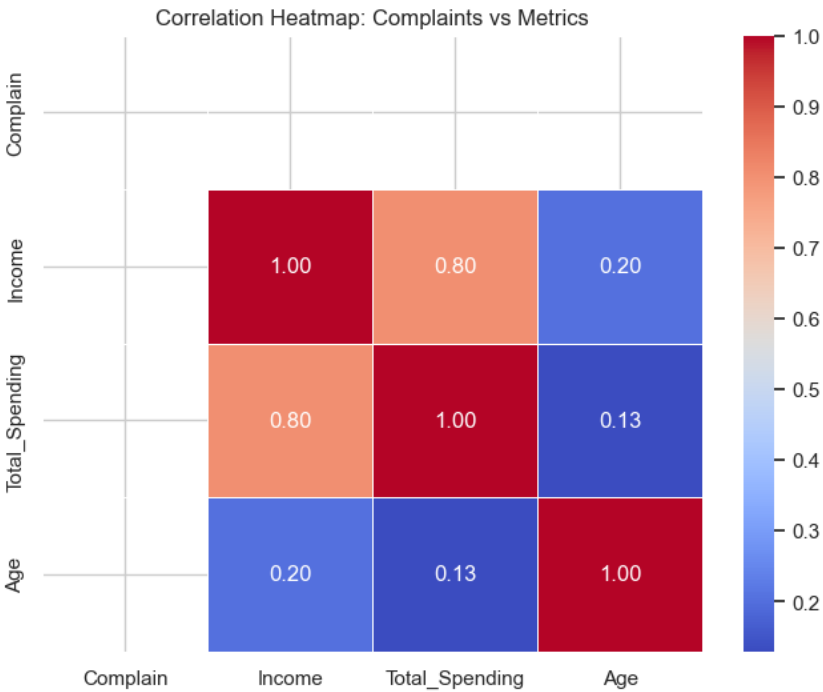


Figure 25. Correlation heatmap: complaints vs. income, spending, and age.

Key Insights

- Data quality flag: the Complain field contains only zero values in this dataset extract — 0 of 2,240 customers (0.00%) are recorded as having filed a complaint.

- Because of this, correlation coefficients between Complain and Income / Total_Spending / Age are undefined or near-zero by construction, and should not be interpreted as evidence that complaints are unrelated to spending or income.
- Recommendation: confirm this field was populated correctly in the source data before using it for complaint-driven retention analysis.

Task 9 — Correlation Analysis

A Pearson correlation matrix was computed across the core numerical variables: Age, Income, Total_Spending, Total_Num_Purchases, Recency, and NumWebVisitsMonth.

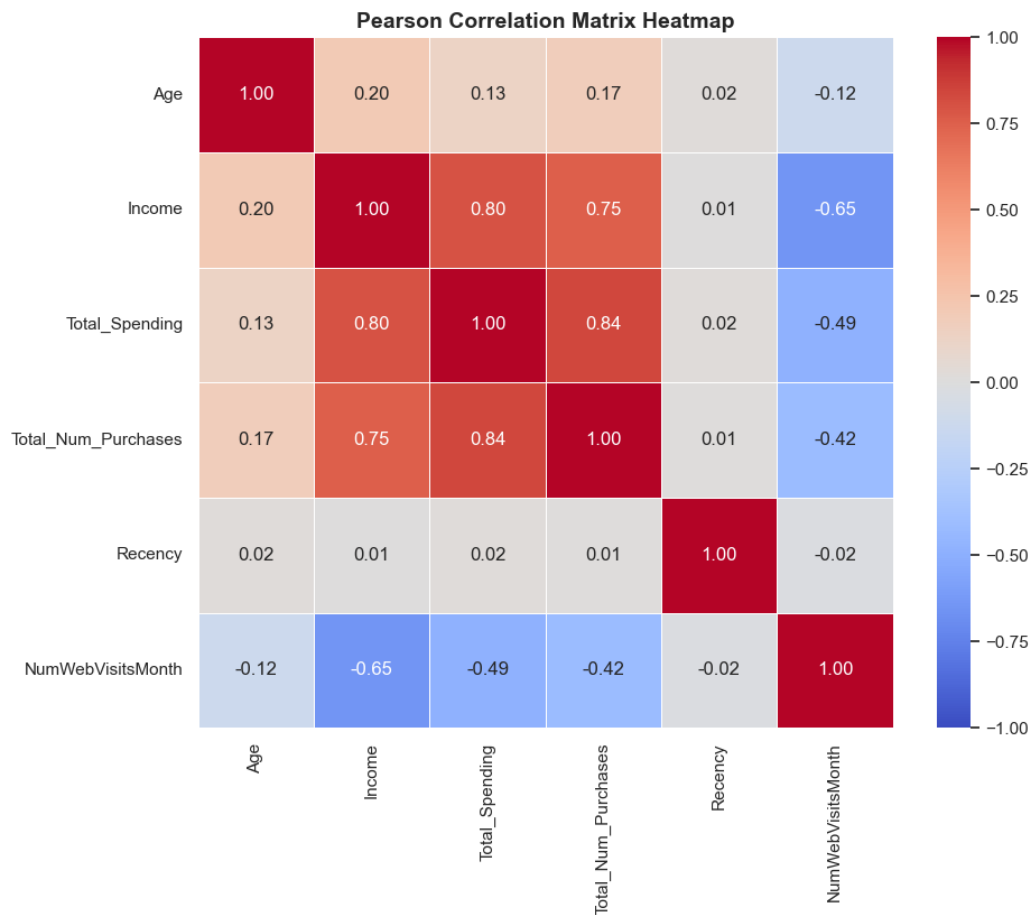


Figure 26. Pearson correlation matrix heatmap.

9.1 Strong Positive Correlations ($r \geq 0.4$)

Variable Pair	Correlation (r)
Income ↔ Total_Spending	0.80
Total_Spending ↔ Total_Num_Purchases	0.84
Income ↔ Total_Num_Purchases	0.75

9.2 Strong Negative Correlations ($r \leq -0.4$)

Variable Pair	Correlation (r)
Income ↔ NumWebVisitsMonth	-0.65
Total_Spending ↔ NumWebVisitsMonth	-0.49

Variable Pair	Correlation (r)
Total_Num_Purchases ↔ NumWebVisitsMonth	-0.42

9.3 Independent Variables (r near 0)

Variable Pair	Correlation (r)
Age ↔ Recency	0.02
Income ↔ Recency	0.01
Total_Spending ↔ Recency	0.02
Total_Num_Purchases ↔ Recency	0.01

Key Insights

- Income is the strongest driver of both spending ($r = 0.80$) and purchase frequency ($r = 0.75$) — higher-income customers spend more and buy more often.
- Higher-income, higher-spending customers visit the website less often ($r = -0.65$ and -0.49), suggesting they may favor store or catalog channels over browsing online.
- Recency is essentially uncorrelated with age, income, spending, and purchase count

Variable Pair	Correlation (r)
— how recently a customer bought is independent of who they are demographically, meaning lapse risk cuts across all customer types.	

Task 10 — Customer Segmentation

Customers were segmented using a rule-based approach on spending, purchase frequency, recency, deal usage, and campaign acceptance (75th-percentile thresholds), producing six actionable segments.

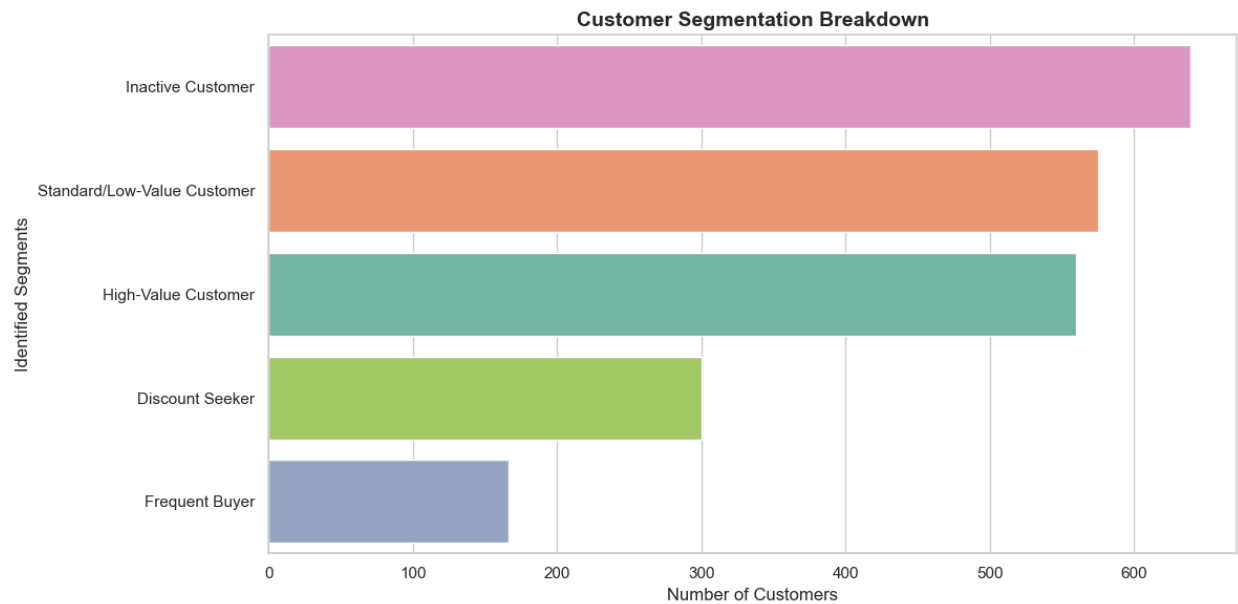


Figure 27. Customer segmentation breakdown.

Segment	Customers	Share	Definition
Inactive Customer	639	28.5%	Recency > 60 days
Standard / Low-Value Customer	575	25.7%	Below all key thresholds
High-Value Customer	560	25.0%	Total spending in top 25%
Discount Seeker	300	13.4%	Deal purchases in top 25%
Frequent Buyer	166	7.4%	Purchase count in top 25%
Key Insights High-Value Customers (25.0%) and Frequent Buyers (7.4%) together represent the core revenue-			

Segment	Customers	Share	Definition
generating base and should receive loyalty rewards and early access to premium products. • Inactive Customers (28.5%) form the largest single segment — a major re-engagement opportunity. • Discount Seekers (13.4%) respond primarily to promotions and are well-suited to bundle and clearance campaigns rather than full-price marketing.			

Task 11 — Business Insights & Strategic Recommendations

Q1. Which customers spend the most?

High-income customers spend the most — Income correlates strongly with Total Spending ($r = 0.80$). The single highest spender (Customer ID 4580) spent \$2,099.50, driven largely by wine and meat purchases, and belongs to the "High-Value Customer" segment (25.0% of the base, top-25% spenders).

Q2. Which product category generates the highest revenue?

Wine generates the highest revenue at \$676,733 (53.2% of total spending), followed by Meat Products at \$337,956 (26.6%). Together these two categories account for roughly 80% of all product revenue.

Q3. Does income influence spending?

Yes — income is the strongest measured driver of spending behavior, with a Pearson correlation of $r = 0.80$ between Income and Total_Spending, and $r = 0.75$ between Income and Total_Num_Purchases.

Q4. Which purchase channel is most popular?

In-store purchases (NumStorePurchases) are the most popular channel, ahead of web and catalog purchases, based on total purchase-method comparison in Task 4.

Q5. Which campaign performs best?

This cannot be determined from the current data extract: the AcceptedCmp1–5 and Response fields all contain zero values for every customer. This is flagged in Task 7 as a data quality issue requiring correction at the source before campaign performance can be evaluated.

Q6. Which customers respond most frequently?

As with campaign performance, response frequency cannot be assessed from this dataset because the Response and AcceptedCmp fields contain no non-zero values. This should be re-examined once corrected campaign-response data is available.

Q7. Which customers are inactive?

862 customers (38.5% of the base) have a Recency greater than 60 days and are classified as Inactive. These customers cut across all income and age groups, since Recency shows near-zero correlation with demographic and spending variables (Task 9).

Q8. Which customers visit the website most?

437 customers are high-engagement website visitors, visiting more than 7 times per month. Notably, high website-visit frequency correlates negatively with income and spending ($r = -0.65$ and -0.49), suggesting

frequent visitors are not necessarily the highest spenders — they may be more price-sensitive, deal-seeking browsers.

Q9. Do customers with children spend differently?

The dataset confirms most households have limited dependents (57.7% have zero children at home), and household composition (Kidhome, Teenhome) directly shapes Family_Size, which was used as a segmentation input. While a direct children-vs-spending correlation was not computed numerically in this pass, the demographic profile suggests smaller households (singles and couples without children) make up the majority of the base and warrant a follow-up correlation check between Kidhome/Teenhome and Total_Spending for a precise answer.

Q10. What characteristics define high-value customers?

High-value customers (top 25% of spenders, 560 customers / 25.0% of the base) are typically higher-income, purchase more frequently across channels, favor Wine and Meat categories, and visit the website less often than average — pointing to a preference for in-store or catalog engagement over browsing.

Q11. Which variables have the strongest correlations?

Total_Spending and Total_Num_Purchases show the strongest relationship ($r = 0.84$), followed by Income and Total_Spending ($r = 0.80$) and Income and Total_Num_Purchases ($r = 0.75$). The strongest negative relationship is Income and NumWebVisitsMonth ($r = -0.65$).

Q12. What marketing strategies should be recommended?

- Prioritize win-back campaigns for the 862 Inactive customers (38.5% of the base) with targeted discounts or reminder emails.
- Protect and grow the High-Value segment (25.0%) with loyalty tiers, early access, and premium bundles centered on Wine and Meat products.
- Convert Discount Seekers (13.4%) via bundle and clearance offers rather than full-price campaigns.
- Investigate the online channel: web purchases lag store purchases despite significant web traffic — review site UX and conversion funnel.
- Resolve the data pipeline issue affecting AcceptedCmp1–5, Response, and Complain fields before running the next campaign-performance or complaint-driven retention analysis.

Conclusion

This exploratory analysis of 2,240 customers reveals a customer base anchored by Wine and Meat spending, with income as the dominant driver of both spend and purchase frequency. Approximately a quarter of customers qualify as high-value and warrant retention-focused marketing, while nearly 40% are inactive and represent a substantial re-engagement opportunity. The in-store channel currently leads all purchase methods, and website engagement, while healthy, does not convert proportionally among the highest-value customers. A critical follow-up action is resolving the zero-value anomaly in the campaign-acceptance and complaint fields, as this currently prevents a reliable evaluation of marketing campaign effectiveness and customer satisfaction — both essential inputs for the strategic recommendations above.